

Customer Segmentation & Churn Prediction Project

Project Description

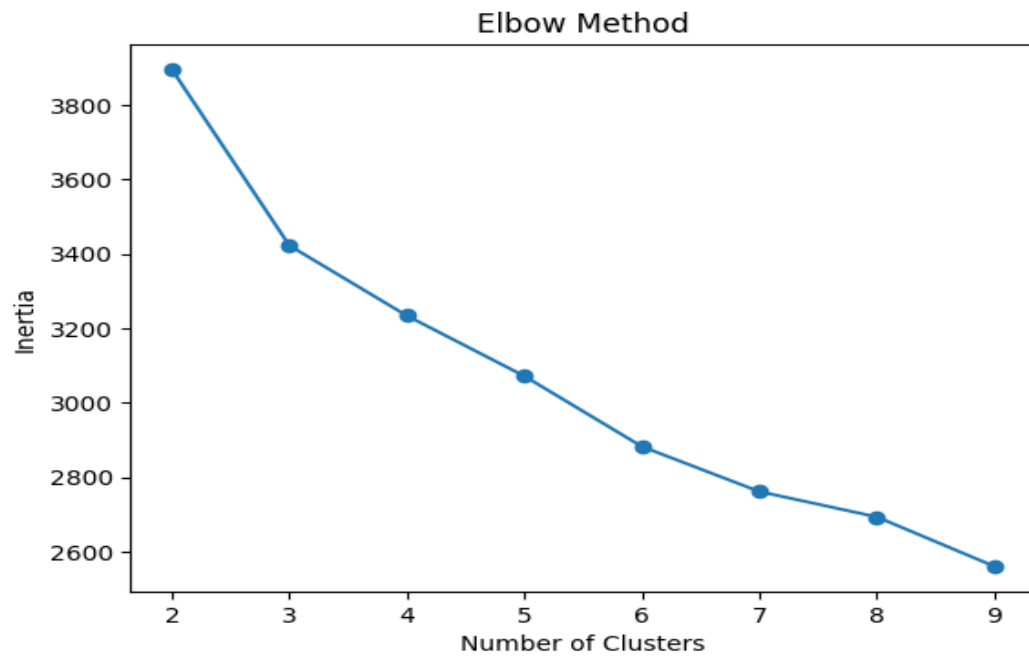
This project focuses on analyzing customer churn data using machine learning techniques. Customers are first segmented using clustering algorithms, and then separate prediction models are built for each segment to accurately predict churn. The goal is to enable businesses to design targeted retention strategies based on customer behavior.

Setup Instructions

1. Install required dependencies using requirements.txt.
2. Open the Jupyter Notebook in Google Colab or local Jupyter environment.
3. Upload the dataset and supporting CSV files.
4. Run cells sequentially to reproduce results.
5. Generated outputs include segmentation data, evaluation metrics, and business insights.

Customer Segmentation Approach

The Elbow Method was used to determine the optimal number of clusters for K-Means clustering. Based on the inertia curve, three clusters were selected as they provide a balance between model simplicity and explanatory power.



Segment Profiling Results

Each cluster was analyzed to understand customer behavior. Key metrics such as tenure, monthly charges, total charges, and churn rate were calculated to form meaningful customer profiles.

```
numeric_profile = df.groupby('KMeans_Cluster').mean(numeric_only=True)
churn_rate = df.groupby('KMeans_Cluster')['Churn'].mean() * 100

segment_profile = numeric_profile.copy()
segment_profile['Churn_Rate_%'] = churn_rate

segment_profile
```

	Tenure	MonthlyCharges	TotalCharges	SeniorCitizen	Churn	Hierarchical_Cluster	DBSCAN_Cluster	Churn_Rate_%
KMeans_Cluster								
0	37.452586	116.737069	4088.387931	0.469828	0.103448	0.025862	0.159483	10.344828
1	37.024194	107.282258	4216.282258	0.524194	0.153226	1.895161	1.459677	15.322581
2	34.625000	114.111111	4497.333333	0.520833	0.069444	0.937500	0.305556	6.944444

Model Evaluation Results

Separate Random Forest models were trained for each customer segment. Performance was evaluated using Accuracy, Precision, Recall, F1-score, and ROC-AUC metrics.



```
results_df = pd.DataFrame(results)
results_df
```

...

	Segment	Accuracy	Precision	Recall	F1	ROC_AUC
0	Premium Spenders	0.935484	0.800000	0.8	0.8	0.984615
1	Budget Conscious	1.000000	1.000000	1.0	1.0	1.000000
2	Young Professionals	0.916667	0.333333	0.5	0.4	0.941176



Next steps:

[Generate code with results_df](#)

[New interactive sheet](#)

```
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Feature Engineering & Encoded Dataset

Categorical variables were converted using one-hot encoding to ensure compatibility with machine learning algorithms. This step ensures consistent feature representation across all segments and models.

	Churn	KMeans_Cluster	Tenure	MonthlyCharges	TotalCharges	SeniorCitizen	Contract_One year	Contract_Two year	PaymentMethod_Credit Card	PaymentMethod_Electronic Check
0	0	1	6	64	1540	1	True	False	True	False
1	0	0	21	113	1753	1	False	False	False	True
2	0	2	27	31	1455	1	False	True	True	False
3	0	0	53	29	7150	1	False	False	False	True
4	0	0	16	185	1023	1	True	False	False	True

Business Recommendations

Premium Spenders: Focus on loyalty programs, premium support, and upselling.

Budget Conscious Customers: Use discounts, bundles, and flexible pricing to reduce churn.

Young Professionals: Emphasize mobile-first design, subscriptions, and referral programs.

By implementing these segment-specific strategies, businesses can significantly improve customer retention and maximize lifetime value.